Dravyavati Corridor Constraint Atlas

Chainage-indexed constraint screening for the proposed elevated corridor along the Dravyavati River, Jaipur · 19 August 2026

Built entirely on open data, independently of JDA and of any consultancy — no project data was used. Every layer in this analysis traces to a publicly accessible URL recorded with an access date, so any of it can be reproduced or contradicted from source.

41.2 km	**413**	**13**	**6**
corridor reconstructed	100 m segments scored	constraint categories	robust hotspot corridors

WHAT THIS IS

A screening tool that flags, every 100 metres along a publicly-reconstructed corridor alignment, where thirteen categories of engineering constraint are likely to bind. It is a triage instrument for deciding where survey effort and design attention should go first. It is not a design document, a hydraulic analysis, a structural assessment, or an alignment recommendation, and it does not substitute for topographical survey, GPR utility mapping, or geotechnical investigation.

METHOD IN FIVE STEPS

1. The channel is taken from OpenStreetMap water polygons and reduced to a centreline via a Voronoi medial axis. **2.** That centreline is cut into 100 m segments keyed on chainage, which is the join key for everything downstream. **3.** Each segment is scored 0–3 against thirteen constraint categories drawn from OpenStreetMap, Google/OSM building footprints, WorldPop, Copernicus GLO-30 elevation and Sentinel-2 imagery. **4.** Scores are combined into an equal-weight composite. **5.** Every constraint weight is then swept ±35% and only chainages whose severity band survives the whole sweep are reported — the rest are artefacts of one weighting choice, not findings.

WHAT THE SCREENING FOUND

Railway crossings	2 distinct crossings
Metro interfaces	6 segments on the operational line, 8 on the line under construction, 25 on proposed alignments
Major arterial crossings	7 distinct crossings
Dam / check structures	21 within 100 m of the alignment
Cross-drainage candidates	23 distinct locations where a modelled tributary meets the corridor
Horizontal curvature	12 segments fall below the IRC:86-2018 minimum radius of 150 m for a 60 km/h design speed
Land and habitation	99.5% of the 60 m corridor buffer is unbuilt on average; approximately 359 buildings lie within 100 m, concentrated in 55 of 413 segments

Robust hotspot corridors

These are the stretches whose severity band did not change under any weighting tested. They are reported because they are stable, not because they are the highest-scoring under our particular assumptions. Contiguous robust segments are grouped by clustering along chainage.

Chainage extent	Length	Composite	Leading constraints
27.9 – 28.4 km	0.5 km	1.04	Existing elevated structure (3.0) · EHT line crossing (3.0) · Built-up growth since 2018 (2.8)
2.4 – 2.7 km	0.3 km	1.03	Metro interface (3.0) · Existing elevated structure (3.0) · Major arterial crossing (3.0)
33.6 – 33.9 km	0.3 km	0.90	EHT line crossing (3.0) · Dam / check structure (3.0) · Hydraulic sensitivity (2.7)
7.9 – 8.1 km	0.2 km	0.88	Existing elevated structure (3.0) · Major arterial crossing (3.0) · EHT line crossing (3.0)
30.5 – 30.8 km	0.3 km	0.87	EHT line crossing (3.0) · Dam / check structure (3.0) · Hydraulic sensitivity (2.3)
41.1 – 41.3 km	0.2 km	0.61	Existing elevated structure (3.0) · Hydraulic sensitivity (3.0) · Entry–exit feasibility (2.0)

Composite is a weighted mean of thirteen 0–3 constraint scores; it is an index for ranking chainages against each other, not a physical quantity.

What drives the ranking

A small neural network was fitted to reproduce the composite from the constraint vector (R^2 0.976), then each constraint was shuffled in turn to measure how far the reproduction degrades. This is a diagnostic of our own index — it does not predict any quantity measured outside this analysis.

Existing elevated structure	0.408
Entry–exit feasibility	0.326
Hydraulic sensitivity	0.306
EHT line crossing	0.281
Metro interface	0.277
Dam / check structure	0.243

The practical reading: proximity to existing elevated structure dominates the ranking, ahead of entry–exit feasibility and hydraulic sensitivity. Habitation proximity — the intuitive first guess for a corridor through a city — ranks well below these.

Confidence, and where this is weakest

Every score in this atlas carries a confidence tag, and the tags are not decoration — they set how far each layer is allowed to move in the uncertainty analysis. Sampling weights and scores jointly over 5,000 runs gives a mean 90% interval width of **0.40** composite points, which is wide relative to the spread between chainages. That is the honest characterisation: this analysis is reliable for ordering and triage, and not for any absolute statement about a single segment.

Power lines (low)	OpenStreetMap coverage of transmission lines in Jaipur is incomplete. EHT crossings flagged here are candidates, not confirmed, and each should be cross-checked against imagery before use.
Restricted / military areas (low)	landuse=military is frequently absent or approximate in India. Where this analysis shows nothing, that is an acknowledged gap in the source data, not evidence that nothing is there.
Built-up growth (low)	Derived from a Sentinel-2 spectral index, not a land-cover classification. Bare soil in this region can mimic built-up signal, so only the 2018-to-2026 change is used, never an absolute.
Hydraulic sensitivity (low)	GLO-30 at 30 m cannot resolve a rectified channel cross-section. The output is a relative screening index over contributing area only. It is not an afflux figure and not a flood claim.
Alignment reconstruction	The corridor is reconstructed from public channel geometry and reaches 41.2 km against a publicly reported ~36 km. A real alignment would ease curves harder than a medial axis can and may bridge across meanders. Chainages here will not correspond exactly to a DPR's.
Land use zoning	The Jaipur Master Plan 2047 land-use layer is absent: the source portal refuses automated requests. No zoning substitute was invented in its place.

Sources

OpenStreetMap via Overpass (channel, rail, metro, roads, power, water structures, buildings; ODbL) · Copernicus DEM GLO-30 via AWS Open Data · Sentinel-2 L2A via Earth Search on AWS · WorldPop India 2020 population density (CC-BY 4.0) · Open-Meteo ERA5 rainfall archive · IRC:86-2018 Table 8.2 for minimum horizontal curve radius · project endpoints and length from published press coverage. Full ledger with access dates and query definitions is in data/SOURCES.md in the repository.

Reproducing and checking this

The pipeline runs end to end from public sources with no credentials and no licensed software. Every intermediate is written to disk, so any number in this brief can be traced back to the layer and the query that produced it. The test suite asserts the invariants that matter — that all spatial analysis happens in a projected CRS, that every score column has a paired confidence column, that the hydraulic index increases downstream, and that the robust hotspot list stays discriminating rather than collapsing to everything or nothing.

Live atlas	https://dravyavati-atlas.vercel.app
Interactive analysis	https://dravyavati-atlas.vercel.app/analysis
Source and data ledger	https://github.com/krish2105/Dravyavati-Project-36

Where this could be wrong

The reconstruction is built from the channel, so it will diverge from a surveyed alignment wherever a designer would depart from the river. Constraint scoring is a proxy exercise: it detects what open data records, and open data under-records exactly the things — buried utilities, land titles, encroachment status — that most often decide an urban corridor. Nothing here has been checked against ground truth, because none was available to this analysis.

> **The ask.** If any of this is useful, I would value twenty minutes to hear where the method is wrong. Correction is more valuable to me than agreement, and the parts most likely to be wrong are the ones listed above.

Screening-grade, not design-grade. Built on open data for constraint triage. All findings require verification against field survey.